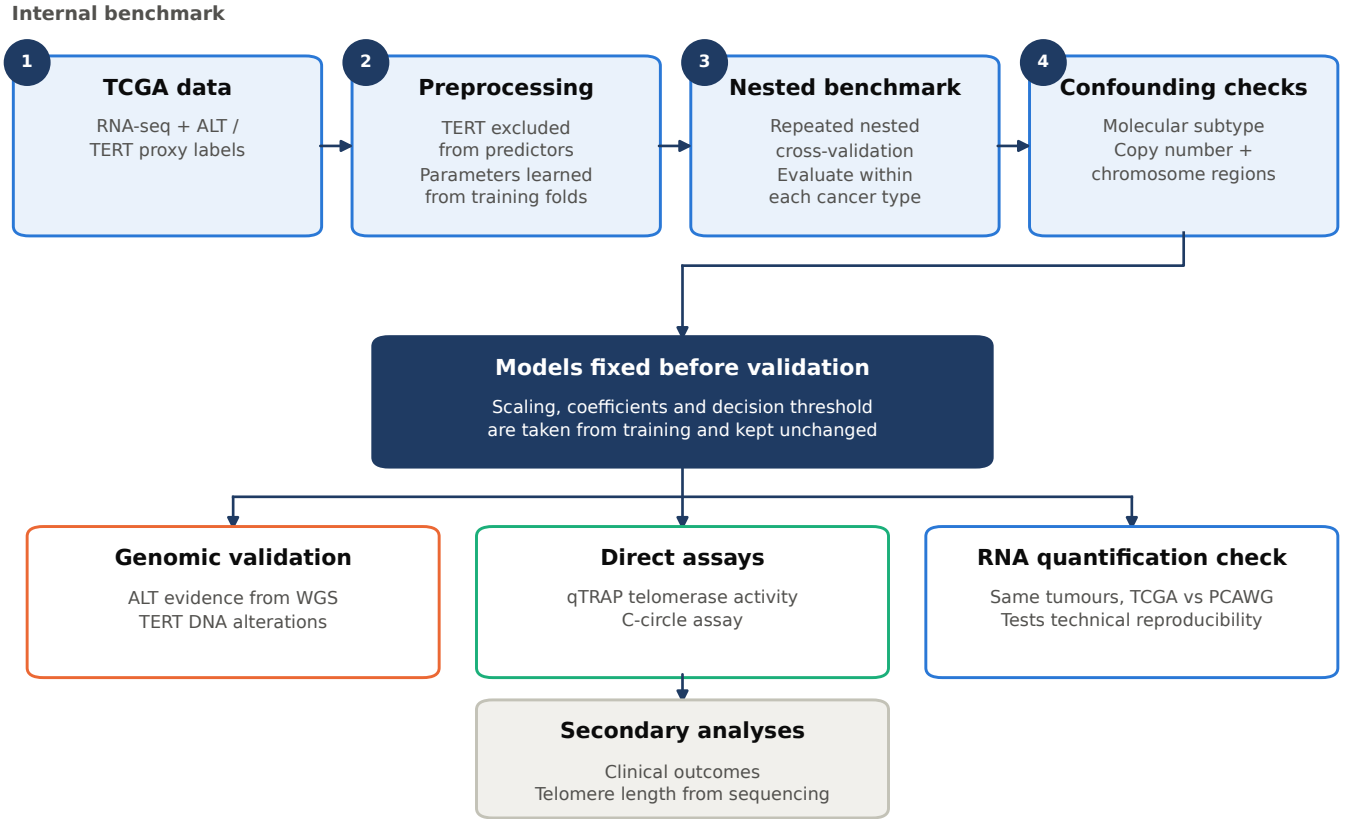

Internal benchmark
1
TCGA data
RNA-seq + ALT / TERT proxy labels
2
Preprocessing
TERT excluded from predictors
Parameters learned from training folds
3
Nested benchmark
Repeated nested cross-validation
Evaluate within each cancer type
4
Confounding checks
Molecular subtype
Copy number + chromosome regions
Models fixed before validation
Scaling, coefficients and decision threshold are taken from training and kept unchanged
Genomic validation
ALT evidence from WGS
TERT DNA alterations
Direct assays
qTRAP telomerase activity
C-circle assay
RNA quantification check
Same tumours, TCGA vs PCAWG
Tests technical reproducibility
Secondary analyses
Clinical outcomes
Telomere length from sequencing